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COVER STORY

The End of Type Annotations

Type annotations are what make statically typed object-oriented languages like Java run faster and safer .

Yegor Bugayenko

Without annotations, every variable would effectively collapse to `Object` , forcing runtime casts. The code would work, but slower and with more runtime errors. Some type annotations may be inferred instead of being explicitly specified by a programmer. Not all though. In Java, for example, a number of hard-to-resolve challenges prevent us from inferring types of all objects. This is not a fundamental limitation of OOP itself. It's a result of design trade-offs in Java and similar languages.

In a perfect object-oriented language, all variable types would be inferrable.

Imagine a simple Java method:

Two reasons justify the usage of `Book` , `City` , `Price` , and `Delivery` type annotations: Compilers and programmers need help.

id="type-annotations-are-helpful">Type Annotations Are Helpful

First, we help the compiler eliminate some dynamic dispatches in favor of static calls. If the `Book` is a class, not an interface, the `book.price()` call may be compiled into a jump to an absolute address. Without information about the book's class, `.price()` goes to a virtual table first, finds the address, and only then jumps. The second scenario is more expensive. The type annotation attached to `book` helps avoid it.

Second, we help ourselves write safe code, avoiding "Method not found" runtime errors . If `book` is not annotated as `Book` , we may mistakenly pass `Integer` , meaning the book's ID in the database. At compile time, that would lead to no errors. Later, at runtime, we get an error when `.price()` is not found in the virtual table of the `Integer` class.

However, both compilers and programmers can improve.

id="compilers-can-do-better">Compilers Can Do Better

Sometimes, a compiler can infer the type of a variable, without an explicit annotation. For example, this code compiles in Java, starting from version 10:

The type annotation used in earlier Java versions is replaced with the `var` keyword.

In a small piece of code such as this one, the compiler can infer types. However, it may fall short with the `book` and `city` parameters. In the general case, type inference is not decidable for Java programs. Because of generics, method overloading, reflection, and ... complexity.

The complexity is the technical obstacle. The compiler can't infer types for all variables because it would be too expensive to analyze the whole program. Instead, it compiles file by file. Even if the compiler had the whole program, inference in Java would still hit undecidability in the general case. File-by-file compilation makes this even more restrictive.

All other barriers, such as generics, the compiler can't overcome:

No matter how hard the compiler tries, in the general case, this question doesn't have an answer.

id="programmers-can-do-better">Programmers Can Do Better

We, programmers, can help the compiler infer types.

For example, we can stop using generics. Instead of List we can have a Library and instead of Map we can have a PhoneBook . It's easier to infer the type of the object taken from a Library versus the object taken from a generic List .

We can also stop using method overloading. Instead of print(String x) and print(Integer x) we can create printString(x) and printInteger(x) . Types of parameters are easier to infer in more specialized methods.

We can also stop using reflection.

The easier it is to infer the type of a variable for a compiler, the faster the programmer gets the semantics of it too.

Java programmers may not be ready for such a radical move . However, if they were, they would not only help the compiler but themselves too. Eliminating type annotations makes code shorter and, because of that, cleaner. This is why var syntax was introduced in Java 10.

In the code above, variable names are nouns. In well-written code, nouns as names are sufficient to disambiguate variables. No need to call it cityOfDelivery or bookToDeliver . Just book and city are enough.

We also named variables by their types: a book is of type Book , and so on. By looking at the name of the variable we can tell its type. The Book type annotation looks like a syntactical redundancy. It only leads to lower code readability, by making it longer. It's reasonable to expect type inference to free our programs from this redundancy.

Thus, better type inference means better readability of the code.

id="languages-can-do-better">Languages Can Do Better

Languages like Haskell and the ML family prove that full type inference is achievable. However, they still need annotations for edge cases. Rust takes a middle ground. It infers local variable types but enforces explicit annotations at public interfaces. Go, until recently, avoided generics, operator overloading, and heavy reflection, making inference straightforward. However, it forces programmers to annotate all public boundaries—function signatures, struct fields, and interfaces.

If the type of a variable can't be inferred, compilation fails.

I suggest taking one step forward and designing a language that doesn't have any type annotations. Such a language should not have generics, method overloading, reflection, and everything else that prevents 100% type inference. Then, we must design a compiler for this language that compiles the entire program, not single files.

The language may be as strict as this: If the type of a variable can't be inferred, compilation fails.

This is how the book price snippet would look in such a language:

To me, this seems to be much more readable than the original code in Java.

Compilation time remains a limitation though. We turn this issue into an opportunity. Programmers are forced, by the timing limitations of the compiler, to write smaller modules. When they need larger programs, they break them into modules that communicate via IPC instead of staying in a monolithic binary.

We're experimenting with this approach in EOLANG , a language designed to maximize inference by eliminating features that make it undecidable.

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FEATURES

Turn Dependabot Off

Filippo Valsorda

Dependabot is a noise machine. It makes you feel like you're doing work, but you're actually discouraging more useful work. This is especially true for security alerts in the Go ecosystem.

I recommend turning it off and replacing it with a pair of scheduled GitHub Actions, one running govulncheck, and the other running your test suite against the latest version of your dependencies.

id="a-little-case-study">A little case study

On Tuesday, I published a security fix for filippo.io/edwards25519 . The (*Point). MultiScalarMult method would produce invalid results if the receiver was not the identity point.

A lot of the Go ecosystem depends on filippo.io/edwards25519, mostly through github.com/go-sql-driver/mysql (228k dependents only on GitHub). Essentially no one uses (*Point). MultiScalarMult .

Yesterday, Dependabot opened thousands of PRs against unaffected repositories to update filippo.io/edwards25519. These PRs were accompanied by a security alert with a nonsensical, made up CVSS v4 score and by a worrying 73% compatibility score , allegedly based on the breakage the update is causing in the ecosystem. Note that the diff between v1.1.0 and v1.1.1 is one line in the method no one uses .

We even got one of these alerts for the Wycheproof repository, which does not import the affected filippo.io/edwards25519 package at all . Instead, it only imports the unaffected filippo.io/edwards25519/field package.

```
$ go mod why -m filippo.io/edwards25519
# filippo.io/edwards25519 github.com/c2sp/wycheproof/
tools/twistcheck filippo.io/edwards25519/field
```

We have turned Dependabot off.

id="use-a-serious-vulnerability-scanner-instead">Use a serious vulnerability scanner instead

But isn't this toil unavoidable, to prevent attackers from exploiting old vulnerabilities in your dependencies? Absolutely not!

Computers are perfectly capable of doing the work of filtering out these irrelevant alerts for you. The Go Vulnerability Database has rich version, package, and symbol metadata for all Go vulnerabilities.

Here's the entry for the filippo.io/edwards25519 vulnerability , also available in standard OSV format .

modules:

- module: filippo.io/edwards25519

versions:

- fixed: 1.1.1

vulnerable_at: 1.1.0

packages:

- package: filippo.io/edwards25519

symbols:

- Point. MultiScalarMult

summary: Invalid result or undefined behavior in filippo.io/edwards25519

description: |-

Previously, if MultiScalarMult was invoked on an

initialized point who was not the identity point, MultiScalarMult

produced an incorrect result. If called on an

uninitialized point, MultiScalarMult exhibited undefined behavior.

cves:

- CVE-2026-26958

credits:

- shaharcohen1
- WeebDataHoarder

references:

- advisory: <https://github.com/FiloSottile/edwards25519/security/advisories/GHSA-fw7p-63qq-7hpr>
- fix: <https://github.com/FiloSottile/edwards25519/commit/d1c650afb95fad0742b98d95f2eb2cf031393abb>

source:

id: go-security-team

created: 2026-02-17T14:45:04.271552-05:00

review_status: REVIEWED

Any decent vulnerability scanner will at the very least filter based on the package, which requires a simple go list -deps ./... . This already silences a lot of noise, because it's common and good practice for modules to separate func-

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Improving the trustworthiness of Javascript on the Web

Michael Rosenberg · Cloudflare Research

The web is the most powerful application platform in existence. As long as you have the right API, you can safely run anything you want in a browser.

Well... anything but cryptography.

It is as true today as it was in 2011 that Javascript cryptography is Considered Harmful. The main problem is code distribution. Consider an end-to-end-encrypted messaging web application. The application generates cryptographic keys in the client's browser that lets users view and send end-to-end encrypted messages to each other. If the application is compromised, what would stop the malicious actor from simply modifying their Javascript to exfiltrate messages?

It is interesting to note that smartphone apps don't have this issue. This is because app stores do a lot of heavy lifting to provide security for the app ecosystem. Specifically, they provide integrity, ensuring that apps being delivered are not tampered with, consistency, ensuring all users get the same app, and transparency, ensuring that the record of versions of an app is truthful and publicly visible.

It would be nice if we could get these properties for our end-to-end encrypted web application, and the web as a whole, without requiring a single central authority like an app store. Further, such a system would benefit all in-browser uses of cryptography, not just end-to-end-encrypted apps. For example, many web-based confidential LLMs, cryptocurrency wallets, and voting systems use in-browser Javascript cryptography for the last step of their verification chains.

In this post, we will provide an early look at such a system, called Web Application Integrity, Consistency, and Transparency (WAICT) that we have helped author. WAICT is a W3C-backed effort among browser vendors, cloud providers, and encrypted communication developers to bring stronger security guarantees to the entire web. We will discuss the problem we need to solve, and build up to a solution resembling the current transparency specification draft. We hope to build even wider consensus on the solution design in the near future.

Defining the Web Application

In order to talk about security guarantees of a web application, it is first necessary to define precisely what the application is. A smartphone application is essentially just a zip file. But a website is made up of interlinked assets, including HTML, Javascript, WASM, and CSS, that can each be locally or externally hosted. Further, if any asset changes, it could drastically change the functioning of the application. A coherent definition of an application thus requires the application to commit to precisely the assets it loads. This is done using integrity features, which we describe now.

An important building block for defining a single coherent application is subresource integrity (SRI). SRI is a feature built into most browsers that permits a website to specify the cryptographic hash of external resources, e.g.,

This causes the browser to fetch underscore.js from cdnjs.cloudflare.com and verify that its SHA-512 hash matches the given hash in the tag. If they match, the script is loaded. If not, an error is thrown and nothing is executed.

If every external script, stylesheet, etc. on a page comes with an SRI integrity attribute, then the whole page is defined by just its HTML. This is close to what we want, but a web application can consist of many pages, and there is no way for a page to enforce the hash of the pages it links to.

We would like to have a way of enforcing integrity on an entire site, i.e., every asset under a domain. For this, WAICT defines an integrity manifest, a configuration file that websites can provide to clients. One important item in the manifest is the asset hashes dictionary, mapping a hash belonging to an asset that the browser might load from that domain, to the path of that asset. Assets that may occur at any path, e.g., an error page, map to the empty string:

```
"hashes": { "81db308d0df59b74d4a9bd25c546f25ec0fdb15a8d6d530c"
"/assets/js/main.js", "fbd1d07879e672fd4557a2fa1bb2e435d88eac072"
"/index.html", "5e737a67c38189a01f73040b06b4a0393b7ea71c86cf73"
"/vendored/main.css", "684ad58287ff2d085927cb1544c7d685ace897b"
"", "f802517f1b2406e308599ca6f4c02d2ae28bb53ff2a5dbcd538391"
"" }
```

The other main component of the manifest is the integrity policy, which tells the browser which data types are being enforced and how strictly. For example, the policy in the manifest below will:

Reject any script before running it, if it's missing an SRI tag and doesn't appear in the hashes

Reject any WASM possibly after running it, if it's missing an SRI tag and doesn't appear in hashes

"integrity-policy": "blocked-destinations=(script), checked-destinations=(wasm)"

Put together, these make up the integrity manifest:

```
"manifest": { "version": 1, "integrity-policy": ..., "hashes":
... }
```

Thus, when both SRI and integrity manifests are used, the entire site and its interpretation by the browser is uniquely determined by the hash of the integrity manifest. This is exactly what we wanted. We have distilled the problem of endowing authenticity, consistent distribution, etc. to a web application to one of endowing the same properties to a single hash.

Recall, a transparent web application is one whose code is stored in a publicly accessible, append-only log. This is helpful in two ways: 1) if a user is served malicious code and

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Technically Speaking - Anniversary Mentoring

Matt Warren (.NET)

I've been reading the excellent Technically Speaking newsletter for a while now and when they announced they would be running a mentoring program, I jumped at the chance and applied straight away. The idea was that each applicant had to set themselves speaking goals or identify areas they wanted to improve and then if you were selected @techspeakingdigest would set you up with a mentor.

I was fortunate enough to be chosen and assigned to Cate one of the authors of the newsletter, who is also a prolific conference speaker. As part of scheme I had to identify the areas that I wanted to improve during the hour-long mentoring session, which for me were:

Turning an outline into a good abstract.

Tips for getting a talk accepted via a CFP submission

I've previously done some talks and they seemed to be well received, but I wanted to expand the range of topics I talked about and try and speak at some other conferences.

id="writing-a-good-abstract">Writing a Good Abstract

At the start of the session Cate looked through an existing submission and offered some advice, which started with the initial comment of:

Good idea, not well pitched

She then went on to offer some really great tips about what conferences were looking for and how I could develop my abstract. I've put the rest of my notes below and left them as I wrote them down, so they are a bit jumbled, but they reflect what happened during the conversation!

id="tips-for-an-abstract-after-reading-mine">Tips for an abstract (after reading mine):

Be pragmatic, too much "one true way" can put people off. Maybe a bit too opinionated.

Don't tie your talk to just one library, might alienate people too much.

id="talk-outlinestructure">Talk outline/structure

Explain - what does it mean to write faster code

Situate - optimisation - what is it? how do you do it? benchmark, etc

Apply - specific examples

id="other-suggestions">Other suggestions

If listeners (or conference organisation committee) agree with your assumptions, they might be more likely to choose your pitch

Be careful about being too specific in the abstract

Don't put too much in the abstract, leave some specifics out

be compelling, but a little bit vague

1 or 2 examples of what not to do is okay, but must give them something to do afterwards, otherwise you could put them off.

Broad v. Narrow talks

Most conferences will want "broader talks"

Bio is pitch for you

Abstract is pitch for you talk

Finally, as well as offering general advice, Cate also took the time to help me re-write an existing abstract I'd put together. I've included the "before" and "after" below, so you can see the difference. Whilst it's hard to see someone pick apart what you're written, I do agree that the "after" reads much better and sounds more compelling than the "before"!

id="before">Before

Microbenchmarks and Optimisations

We all want to write faster code right, but how do we know it really is faster, how do we measure it correctly?

During this talk we will look at what mistakes to avoid when benchmarking. .NET code and how to do it accurately. Along the way we will also discover some surprising code optimisations and explore why they are happening

id="after">After

Where the Wild Things Are - Finding Performance Problems Before They Bite You

You don't want to prematurely optimize, but sometimes you want to optimize, the question is - where to start? Benchmarking can help you figure out what your application is doing and where performance problems could arise - allowing you to find (and fix!) them before your customers do.

If you aren't already benchmarking your code this talk will offer some starting points. We'll look at how to accurately benchmark in .NET and things to avoid. Along the way we'll also discover some surprising code optimisations!

id="the-end-result">The End Result

After the mentoring with Cate took place I was accepted to talk at ProgSCon London 2016, so obviously the tips and re-write of my abstract made a big difference!!

So thanks to Chiu-Ki Chan and Cate for producing Technically Speaking every week, it's certainly helped me out!

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standard-article(title: [Nova Hackathon], author: [firiaati], source-name: [Finn.no Tech], images: (), paragraphs: ([A Hackathon is period of time where computer programmers collaborate intensively on a given task or multiple tasks. For the Nova Hackathon Challenge Finn was invited to make a task for the students along with 2 other companies.], [The challenge that Finn gave the students was to setup a simple web-server. It was based on Extreme startup by Robert Chatley . This software supports a workshop where teams can compete to build a software product that satisfies market demand.], [We helped the students to setup their web-servers, each in their own respective language of choice. Their web-server's url had to be registered on the Finn server. Once this was setup correctly, the Finn server would start to send http requests to their web-server. These http requests were various questions, the students had to get their web-server to respond to these requests not knowing the questions in advance, nor how they would be scored. The only thing they knew was that they would get a http request from the FINN server and had to respond to it. The questions were sent at a quick pace and were constantly evolving. This meant that their web-server's code also was quickly evolving, a condensed real-life example of team-based agile programming.], [Avanade's task was to build an application for retrieving relevant data for a transport company. The students would then setup an prototype and a suggestion for further improvements. Itera gave the students a task centered around a calculator for mortgages, with focus on front-end javascript. It has to be mentioned that HP was so kind to borrow their offices for the Hackathon event. Read what HP has written about the Hackaton event .], [The students was divided into 4 groups with different backgrounds. The business representatives helped the students underway and as you can see by this picture], [the students was highly focused and competitive against the other groups.], [Feedback from Nova was that no company in any of their events had ever been so involved throughout the whole day with the students as we had been. We had heaps of fun together and the extreme startup competition was a fantastic suggestion, hands-on and exciting. We had one team who implemented it in php and finished all 8 rounds in under 2 hours including questions that involved html scraping finn.no to find titles and prices for random finnkod! It was fun to see all the hard work and that the students put a tremendous effort in the challenge. The Finn ambassadors, Anders Olsen Sandvik , Mick Semb Sever , Nicolai Høge , Sverre Roos and Rida Aatif had a lot of fun collaborating with the students and we can agree that this is something we would definitely do again.],), insert-map: (:), word-count: 465, edited-for-length: false, debug-mode: false,)

standard-article(title: [Convert An Uploaded Image To A Base64 String In Node.js], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)], images: (), paragraphs: ([Not too long ago I wrote about uploading files to a Node.js server that was making use of the Multer middleware. In that scenario the files were saved to the file system and even though I didn't demonstrate it, in theory the file metadata would be stored in a database. What if you wanted to store the files in a database as a base64 string instead of on the file system?], [We're going to take a look at what it would take to allow file uploads in Node.js and convert them into base64 strings for database storage.], [The post Convert An Uploaded Image To A Base64 String In Node.js appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 116, edited-for-length: false, debug-mode: false,)

The questions were sent at a quick pace and were constantly evolving.

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standard-article(title: [Migrating Your Content Management System (CMS) Assets With MongoDB and Node.js], author: [Nic Raboy], source-name: [Nic Raboy (polyglot

standard-article(title: [Backup WordPress Database And Filesystem Data On Linux With Scripts], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)],

developer)], images: (), paragraphs: ([Content platforms evolve as business strategies shift. At MongoDB, we embraced external publishing platforms like Dev.to , Medium , The Polyglot Developer , etc. to better engage developer communities, requiring us to redistribute content while maintaining our existing CMS data in MongoDB.], [To support our multi-platform publishing strategy, we created a system to publish content between our MongoDB CMS and external platforms. As a result, we needed to migrate the content we had in our CMS to its new home. The migration process included exporting the written content stored in MongoDB and downloading a copy of the media assets that were stored on third-party servers.], [In this tutorial, we'll explore the export process to get the job done with as little friction as possible.], [The post Migrating Your Content Management System (CMS) Assets With MongoDB and Node.js appeared first on DEV .],), insert-map: (:), word-count: 142, edited-for-length: false, debug-mode: false,)

standard-article(title: [Implementing U2F Authentication With Hardware Keys Using Node.js and Vue.js], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)], images: (), paragraphs: ([Not too long ago I had written a tutorial titled, U2F Authentication with a YubiKey Using Node.js and jQuery , which demonstrated how to use hardware keys as a means of universal two-factor (U2F) authentication. However, I had left some things to be desired in that previous post. For example, the previous tutorial did not use proper session management with Express.js and it used jQuery, which is neat, but by no means is as popular as some of the other web frameworks that currently exist.], [In this tutorial, we're going to expand upon what we had done previously, but implement proper session management with Express.js and use Vue.js , which is a modern web framework.], [The post Implementing U2F Authentication With Hardware Keys Using Node.js and Vue.js appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 134, edited-for-length: false, debug-mode: false,)

standard-article(title: [Display Toast Notifications In A NativeScript Angular Application], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)], images: (), paragraphs: ([Back when I had just started working with NativeScript, about a year ago, I wrote a tutorial for displaying Toast notifications in Android . While still valid, it demonstrated these notifications using vanilla NativeScript and JavaScript. Not only that, but the notifications were only for Android. Since the framework has become significantly better, I figured it would be a good idea to demonstrate these Toast notifications in a NativeScript Angular application.], [We're going to see how to use a plugin to display Toast notifications in a NativeScript Android and iOS application built with Angular.], [The post Display Toast Notifications In A NativeScript Angular Application appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 111, edited-for-length: false, debug-mode: false,)

standard-article(title: [Make Your Own Facebook Mobile App With Ionic Framework], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)], images: (), paragraphs: ([Not too long ago I contributed to the ng-cordova

images: (), paragraphs: ([If you're like me, you run a WordPress blog and are terrified of the thought of something going wrong. With core updates, theme updates, plugin updates, and server component updates, there is a lot of room for error. This is where a WordPress backup could help ease your mind.], [WordPress recommends taking a backup of your blog before any of these are done and there are even some popular plugins that will do this for you. For example, you could use the popular UpdraftPlus or similar, but I believe there is room for error in those as well. While I could be wrong, I think WordPress must be in good shape for backup plugins to be successful.], [The alternative would be to create your own backup scripts that run on a cron schedule. We're going to see how to do this for WordPress instances running on a Linux machine.], [The post Backup WordPress Database And Filesystem Data On Linux With Scripts appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 168, edited-for-length: false, debug-mode: false,)

standard-article(title: [Query A GraphQL API With Vue.js And Axios], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)], images: (), paragraphs: ([I've written quite a bit of content around developing a GraphQL API with various programming technologies such as Node.js, Golang, and Java. Heck, I even wrote an eBook on the subject titled, Web Services for the JavaScript Developer . However, I haven't really produced any content around interacting with those APIs using modern frameworks and client facing technologies, only cURL and Postman.], [So how do you interact with a GraphQL API using something like Angular, React, or Vue.js?], [There are a lot of frameworks to cover, but in this particular tutorial we're going to see how to use Vue.js and simple JavaScript to interact with a GraphQL API.], [The post Query A GraphQL API With Vue.js And Axios appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 125, edited-for-length: false, debug-mode: false,)

standard-article(title: [Upcoming Presentation: Couchbase Live NYC 2015], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)], images: (), paragraphs: ([I'm trying something new so bare with me here. I speak at quite a few events (conferences, meetups, code camps, etc.) so I figured it would be useful to my followers to know when and where these events are in case I am in your part of the world.], [I'll keep it short and to the point. I'm going to be in New York on October 6th, 2015 to present at Couchbase Live New York . It is free to attend this conference, so if you're in the area, please swing by.], [My presentation will be on Node.js and is titled Getting Started: Couchbase Server Node.js SDK .], [The abstract to my talk is as follows:], [The post Upcoming Presentation: Couchbase Live NYC 2015 appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 131, edited-for-length: false, debug-mode: false,)

standard-article(title: [Extract the Version Information of a Game with Unity and C#], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)], images: (), paragraphs: ([If you play a lot of games, you probably noticed

project on GitHub. I added Oauth functionality to many popular API providers such as Google, Facebook, and Twitter.], [I made a guide on how to use this added Oauth functionality on the official Ionic Framework blog , but since then I've received a good deal of requests to take it a step further and actually use on of the APIs beyond just Oauth.], [This tutorial will help you better understand how to use the Facebook APIs for use in your Phonegap or Ionic Framework mobile application.], [The post Make Your Own Facebook Mobile App With Ionic Framework appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 113, edited-for-length: false, debug-mode: false,)

standard-article(title: [Using Google Admob In Your NativeScript Angular Mobile App], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)], images: (), paragraphs: ([I recently received a request from one of my followers for a tutorial on using Google Admob in a NativeScript Angular application. Not too long ago I had demonstrated Google Admob in a vanilla NativeScript project , but I hadn't yet given it a shot with Angular.], [In case you're unfamiliar with Admob, it is an excellent way to monetize your mobile applications with advertisements. You'll earn revenue not only from people clicking your in-app advertisements, but also from the advertisements appearing on the screen.], [We're going to see how to use Admob in a NativeScript Android and iOS application that was built with Angular.], [The post Using Google Admob In Your NativeScript Angular Mobile App appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 123, edited-for-length: false, debug-mode: false,)

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ANALYSIS

at some point in time that the version number or build number of the game is often presented clearly on the screen. This is often not by accident and is instead a way to show users that they are using the correct version of your game. Not only that, but it can help from a development perspective as well. For example, have you ever built a game and had a bunch of different builds floating around on your computer? Imagine trying to figure out which build is correct without seeing any build or version information!], [In this tutorial we're going to see how to very easily extract the build information defined in the project settings of a Unity game.], [The post Extract the Version Information of a Game with Unity and C# appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 149, edited-for-length: false, debug-mode: false,)

standard-article(title: [TPDP Episode #7: Developing For The Internet Of Things (IoT) And Where To Start], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)], images: (), paragraphs: ([When it comes to software development there are many different items to develop for. I've spent most of my developer life creating applications for the web, desktop, and mobile, but when it comes to possibilities it doesn't end there. In recent years the Internet of Things (IoT) has been taking over. In this episode of The Polyglot Developer Podcast, guest speaker Jen Looper and I discuss the Internet of Things and what you need to get started as a developer.], [Jen Looper is a developer advocate at Progress Software, but is also a huge advocate for IoT. In Episode #7: Developing for The Internet of Things (IoT) and Where to Start , we're going to discuss quite a bit, covering everything from a background on IoT to possible development technologies for various hardware.], [The post TPDP Episode #7: Developing For The Internet Of Things (IoT) And Where To Start appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 156, edited-for-length: false, debug-mode: false,)

When flying from Oslo to London, you don't want to go via Moscow

Per Jørgen Walstrøm · Finn.no Tech

If you want to travel from Oslo to London, you may end up getting results which include layovers in far-away places. Do you really want to spend a night at the airport in Moscow if you only want to go from Oslo to London? Probably not, but nonetheless, some of the results we receive at the FINN Flight Search are those kind of travels. Those travels may be really cheap, ending up at the top of our result list as we sort by price, but they are mostly garbage and we do not want our users to be confused by those outliers. We want them to go away!

id="the-old-solution">The old solution

We have tried to fix this before. With limited success. A solution we used a couple of years back was something like this:

We removed results with duration > x. This works fine on short travels, like a two-hour flight from Oslo to London. With the given formula, all trips longer than 6 hours to London will be removed. That's ok.

However, it doesn't work too well on longer flights. If you want to go from Oslo to Bangkok, you would probably spend at least 11 hours in the air. In that case, it would be too conservative to just remove the flights that are 33 hours or more.

id="a-theoretical-solution">A theoretical solution

The correct way to remove the outliers is by finding the quartiles. When the quartiles are found, the upper fence is defined by

Where IQR is the Interquartile Range; Q3 - Q1.

We are using Solr at FINN, but the stats query in Solr does not give us the quartiles out of the box, so we would need to do a separate query to calculate the quartiles. Should be easy enough.

id="what-is-too-long">What is too long?

So what is a too long flight? That's not an easy question to answer. Racking our brains (and doing some guesswork), we came up with the following suggestions on what the upper fence on duration to different destinations should be. The durations are per leg:

class="highlight"> OSL - LON 300 mins = 5 hrs. Given the minimum flight time OSL - LON = 115 OSL - BKK (Bangkok) 1000 mins = 17 hrs. Minimum = 674 OSL - KOA (Kona, Hawaii) 2000 mins = 33 hrs. Minimum = 1335

The minimum flight duration is clearly of interest to us. We store that away in Solr, and using the stats query from Solr on that field, we get the following data on a flight OSL-LON:

A lot of good information there! We get the standard deviation, mean and min values. Missing the quartiles, though, but we can fix that!

id="and-what-is-too-short">And what is too short?

Sometimes, the supplier sends us incorrect data. That's bad. But the good thing is that when the data is incorrect it is really incorrect. We sometimes receive flight times of 0 minutes. That is obviously wrong, and we filter out those. However, we have not seen any occurrences of, say, 1 minute flight times. That would be wrong as well, but more difficult to handle. So we close our eyes and don't care about those.

id="but-hang-on">But hang on...

Minimum = 674 OSL - KOA (Kona, Hawaii) 2000 mins = 33 hrs.

Per Jørgen Walstrøm

So, we were ready to go! We wanted to fix this problem once and for all. But before we started coding, we talked to emeritus Harald Goldstein at the University of Oslo. He's been teaching statistics at the university level for years and had some good feedback on our theoretical solution. First of all, he pointed out that the solution we have outlined is a good solution if we have a symmetrical distribution of the flight durations. The quartiles and the standard deviation works well on a symmetrical distribution like the normal distribution, but what about our data? Is it normal distributed? That was a relevant question, and as one of FINN's main strategies is "data in our backbones", we surely needed to have a closer look at this.

So here goes, flights from Oslo to London. Trip duration on the x-axis, number of occurrences on the y-axis:

The shortest trip duration OSL - LON is 230 minutes in the graph above (115 min per leg) and there is even an outlier - which is not visible - at 2750 minutes! However, it is quite clear that this is not a normal distribution. It might look more like a gamma distribution for the part where $x > 230$.

Darn!

id="back-to-square-one">Back to square one

All the fancy statistics didn't provide the solution, but it helped us really understand the problem. Obviously, we needed to re-think.

id="logarithm">Logarithm

A problem with the solution from ages back (described at the beginning), was that it didn't work very well on long hauls. The relation between the minimum trip duration OSL-BKK and OSL-LON is approx. 6:1. Alas, the upper fence must be relatively lower on longer trips. After some more brain-racking, we came up with the idea that we should look at this logarithmically.

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EOF is not a character

Ruslan Spivak

Update Mar 14, 2020 : I'm working on an update to the article based on all the feedback I've received so far. Stay tuned!

I was reading Computer Systems: A Programmer's Perspective the other day and in the chapter on Unix I/O the authors mention that there is no explicit "EOF character" at the end of a file .

If you've spent some time reading and/or playing with Unix I/O and have written some C programs that read text files and run on Unix/Linux, that statement is probably obvious. But let's take a closer look at the following two points related to the statement in the book:

EOF is not a character

EOF is not a character you find at the end of a file

1. Why would anyone say or think that EOF is a character? I think it may be because in some C programs you can find code that explicitly checks for EOF using `getchar()` and `getc()` routines:

OR

```
FILE * fp ; int c ; ... while (( c = getc ( fp )) != EOF ) putc ( c , stdout );
```

And if you check the man page for `getchar()` or `getc()` , you'll read that both routines get the next character from the input stream. So that could be what leads to a confusion about the nature of EOF , but that's just me speculating. Let's get back to the point that EOF is not a character.

What is a character anyway? A character is the smallest component of a text. 'A', 'a', 'B', 'b' are all different characters. A character has a numeric value that is called a code point in the Unicode standard. For example, the English character 'A' has a numeric value of 65 in decimal. You can check this quickly in a Python shell:

Or you could look it up in the ASCII table on your Unix/Linux box:

Let's check the value of EOF by writing a little C program. In ANSI C, EOF is defined in as part of the standard library. Its value is usually -1. Save the following code in file `printeof.c` , compile it, and run it:

```
int main ( int argc , char * argv []) { printf ( "EOF value on my system: %d\n" , EOF ); return 0 ; }
```

```
$ ./printeof EOF value on my system: -1
```

Okay, so on my system the value is -1 (I tested it both on Mac OS and Ubuntu Linux). Is there a character with a numerical value of -1? Again, you could check the available numeric values in the ASCII table or check the official Unicode page

to find the legitimate range of numeric values for representing characters. But let's fire up a Python shell and use the built-in `chr()` function to return a character for -1:

```
class="highlight">$ python >>> chr ( -1 ) Traceback ( most recent call last ) : File " " , line 1 , in ValueError: chr () arg not in range ( 0x110000 )
```

As expected, there is no character with a numeric value of -1. Okay, so EOF (as seen in C programs) is not a character.

Onto the second point.

2. Is EOF a character that you can find at the end of a file? I think at this point you already know the answer, but let's double check our assumption.

Let's take a simple text file `helloworld.txt` and get a hexdump of the contents of the file. We can use `xxd` for that:

```
$ xxd helloworld.txt 00000000 : 4865 6c6c 6f20 776f 726c 6421 0a Hello world!.
```

As you can see, the last character at the end of the file is the hex `0a` . You can find in the ASCII table that `0a` represents `nl`, the newline character. Or you can check it in a Python shell:

Okay. If EOF is not a character and it's not a character that you find at the end of a file, what is it then?

EOF (end-of-file) is a condition provided by the kernel that can be detected by an application.

Let's see how we can detect the EOF condition in various programming languages when reading a text file using high-level I/O routines provided by the languages. For this purpose, we'll write a very simple cat version called `mcats` that reads an ASCII -encoded text file byte by byte (character by character) and explicitly checks for EOF . Let's write our cat version in the following programming languages:

ANSI C

JavaScript (node.js)

You can find source code for all of the examples in this article on GitHub . Okay, let's get started with the venerable C programming language.

ANSI C (a modified cat version from The C Programming Language book)

```
int main ( int argc , char * argv []) { FILE * fp ; int c ;
```

```
if (( fp = fopen ( *++ argv , "r" )) == NULL ) { printf ( "mcats: can't open %s\n" , * argv ); return 1 ; }
```

```
while (( c = getc ( fp )) != EOF ) putc ( c , stdout );
```

```
fclose ( fp );
```

```
return 0 ; }
```

}

standard-article(title: [Navigating A Web Application With The Angular Router], author: [Nic Raboy], source-name: [Nic Raboy (polyglot developer)], images: (), paragraphs: ([I've been keeping up with Angular since the beta releases and if you have too you'll know that navigation has changed about one hundred times between then and the now stable release. Navigation with the Angular Router component is a tricky subject, but understanding it is necessary for pretty much every quality Angular web application.], [I wrote a now obsolete tutorial on how to navigate between Angular routes back when Angular was in beta. Since Angular is now stable, I thought it would be a good idea to share how to navigate between pages with the stable Angular Router component.], [The post Navigating A Web Application With The Angular Router appeared first on The Polyglot Developer.],), insert-map: (:), word-count: 117, edited-for-length: false, debug-mode: false,)

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download. The episode titled, Going FaaS on Serverless with OpenWhisk focuses on the development and deployment of functions to be deployed on a serverless cloud platform.], [In this episode I'm joined by Carlos Santana who is part of the Cloud Functions team at IBM. While Apache OpenWhisk will have an emphasis in this episode, other cloud services are mentioned as well, including AWS Lambda and Azure Functions.], [The post TPDP Episode #17: Going FaaS On Serverless With OpenWhisk appeared first on The Polyglot Developer .],), insert-map: (:), word-count: 103, edited-for-length: false, debug-mode: false,)

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Find The Application Version In Your Ionic Framework App

Nic Raboy · Nic Raboy (polyglot developer)

If you're like me, you want your users to know which version of your mobile application they are using. Nothing like feeling several generations behind without even knowing it. There are other benefits to knowing your application version. For example, maybe you need to upgrade legacy data when people upgrade between versions of your application. You can't assume your users will be upgrading from the latest version of your application.

Now you can always hard-code the application version throughout your application and hope you remember to change it upon a new release, or you can make use of the nifty Apache Cordova plugin from White October .

My current favorite Apache Cordova framework is Ionic , so we'll be configuring based on that. However, it is not difficult to use this information with Phonegap .

The post Find The Application Version In Your Ionic Framework App appeared first on The Polyglot Developer .

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how to throttle specific requests in Chrome DevTools, and more.], [► Watch on YouTube], [Find out why you should use self-destructing CSS pattern when working with web components, how to throttle specific requests in Chrome DevTools, and more.], [Free Your Code Time: Get 50% Off Managed Hosting & 50 Free Migrations], [00:00 Intro 00:24 Enhancing Web Components Safely with Self-Destructing CSS 01:49 Cloudways — Get 50% Off Managed Hosting 02:32 How To Throttle Specific Requests In Chrome DevTools 03:58 It is OK to Say “CSS Variables” Instead of (or Alongside) “Custom Properties” 04:49 SVG Gobbler 05:39 Choose Your Monster - Customizable Select],), insert-map: (:), word-count: 117, edited-for-length: false, debug-mode: false,)

standard-article(title: [Automate Your Testing Workflows Using BrowserStack's AI Agents], author: [Zoran Jambor], source-name: [CSS Weekly], images: (), paragraphs: ([Find out how to use BrowserStack's AI agents to automate your testing workflows.], [► Watch on YouTube], [In this hands-on guide, you'll see how to easily use the Test Case Generator Agent to turn user stories into detailed, structured test cases in seconds, and then convert those cases into automated tests with the Low-Code Authoring Agent.], [Try BrowserStack for free], [Baseline Status for Video], [00:00 Why you should automate your testing workflows 00:58 How to opt-in to BrowserStack AI 01:36 Overview of the demo project 03:09 Overview of BrowserStack AI Agents 03:31 How to use the Test Case Generator Agent to create structured test cases 06:03 How to use the Low-Code Authoring Agent to create automated tests 09:10 How to record a test flow using the Low-Code Automation 10:03 How to use AI to control recording flow in the Low-Code Automation 12:14 Why you should start using BrowserStack AI today],), insert-map: (:), word-count: 151, edited-for-length: false, debug-mode: false,)

Robin Ward (eviltrout), Robin Ward (eviltrout): I have two 4K TVs at home, one in the bedroom and one in my living room. Both have consoles attached. One has a PS4 Pro and one has a Nintendo Switch and both are great.

However, I also have a gaming PC in my office, which is quite powerful compared to any home console Mine is equipped with a RTX 2080 TI GPU, and a 9900K processor, so it can play pretty much any new game at 4K with a solid 60 fps on high settings.

The Hacker News, The Hacker News (cybersecurity): Cybersecurity researchers have flagged a new evolution of the GlassWorm campaign that delivers a multi-stage framework capable of comprehensive data theft and installing a remote access trojan (RAT), which deploys an information-stealing Google Chrome extension masquerading as an offline version of Google Docs. “It logs keystrokes, dumps cookies and session tokens, captures screenshots, and

The Hacker News, The Hacker News (cybersecurity): Cybersecurity researchers have disclosed a vulnerability in Anthropic's Claude Google Chrome Extension that could have been exploited to trigger malicious prompts simply by visiting a web page. The flaw “allowed any website to silently inject prompts into that assistant as if the user wrote them,” Koi Security researcher Oren Yomtov said in a report shared with The Hacker News. “No clicks, no

Nic Raboy, Nic Raboy (polyglot developer): As an Android developer, you’ll often run into the scenario where you need to perform tasks and display notifications for your app in the background.

To retain battery power on our users device we are going to run background tasks using a broadcast receiver. This will prevent a thread from constantly running in the background draining the battery quickly over time. Instead, the task will be run on defined intervals of the alarm.

The post Use A Broadcast Receiver For Background Services In Android appeared first on The Polyglot Developer .

Parth Pathak,Julie Rudd,Damian Pfister, Elastic Blog: We use AI and RAG to accelerate answers — not replace experts. Every response is reviewed, validated, and refined by engineers to ensure accurate, high-quality, expert-verified solutions, never automated output.

Nic Raboy, Nic Raboy (polyglot developer): This article was designed to help you apply your knowledge of AngularJS and Ionic Framework and create a functional app rather than learn new concepts.

This tutorial should help you with the following concepts:

Make use of the AngularJS \$http service for handling HTTP requests

Cache HTTP response data on the device for usage offline

Create data lists in Ionic Framework

Open external URLs with the Apache Cordova InAppBrowser plugin

The post Create An RSS Reader Using AngularJS And Ionic Framework appeared first on The Polyglot Developer .

Karen Mcdermott, Elastic Blog: Leading financial companies are turning data ubiquity into competitive advantage. See how Elastic powers contextual search, real-time decisioning, and AI agents across fraud, compliance, and customer experience.

Karen Mcdermott, Elastic Blog: From AI-powered insight to stablecoin innovation, Money20/20 USA 2025 showed that fintech’s future is built on unified data, real-time intelligence, and the infrastructure of trust.

Alf Franklin, Elastic Blog: Data sovereignty is central to defence strategy, allowing organisations to maintain data control, protect sensitive information, act confidently, and ensure mission readiness.

Nic Raboy, Nic Raboy (polyglot developer): Previously I wrote about using the device camera in your React Native mobile application. However, what if we wanted to go a step further and start scanning barcodes?

This is not a problem because the react-native-camera component we used in the last tutorial also supports barcode scanning.

The post Implement A Barcode Scanner Using React Native appeared first on The Polyglot Developer .

Jesse Orrall, CNET News: Quantum computers promise to revolutionize whole industries by outperforming classical computers on complex calculations. They just need to be colder than the coldest natural place in the universe.

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